Document Technique

Pipeline JDBC avec Joins, Nettoyage et Detection d'Anomalies

Source : ERP Ask&Go — Multi-tables avec jointures

companies x invoices x invoice_lines x payments

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Table des matieres

1. Schema ERP (Ask&Go)

```
+---------------+    +---------------+    +---------------+
|  companies    |    |   invoices    |    | invoice_lines |
|  (societes)   |    |  (factures)   |    |   (lignes)    |
+---------------+    +---------------+    +---------------+
| company_id PK |<---| company_id FK |    | invoice_id FK |
| tenant_code   |    | invoice_id PK |<---|  line_id PK   |
| name          |    | invoice_date  |    |    amount     |
| sector        |    |    status     |    |    label      |
|  is_active    |    |   doc_type    |    |   category    |
+---------------+    |   currency    |    +---------------+
                     +---------------+
                           |
                           v
                     +---------------+
                     |   payments    |
                     |  (paiements)  |
                     +---------------+
                     | payment_id PK |
                     | invoice_id FK |
                     |  paid_amount  |
                     | payment_date  |
                     |    method     |
                     +---------------+
```

Le schema comporte 4 tables reliees par des cles etrangeres. Le pipeline JDBC effectue des jointures `INNER JOIN` entre `invoices`, `companies` et `invoice_lines`, puis applique des filtres multi-criteres via `PreparedStatement`.

2. Donnees d'exemple (avec anomalies injectees)

2.1 Table companies

Tableau 1 Table companies — donnees source

company_id	tenant_code	name	sector	is_active
C001	whitecape_ask	EAU_SAISON	Utilities	true
C002	whitecape_ask	ELEC_HEBDO	Utilities	true
C003	whitecape_ask	INTERNET_FIBRE	Telecom	true
C004	whitecape_ask	FOURNITURES_BUR	Office	true
C005	whitecape_ask	CONSULTING_IT	Services	true
C006	whitecape_ask	**NULL**	NULL	true
C007	other_tenant	SECRET_CORP	Finance	true

> ***Anomalies :*** *C006 a un nom NULL (sera rejete a la normalisation). C007 a un mauvais tenant (sera filtre par la condition SQL* `tenant_code = 'whitecape_ask'`*).*

2.2 Table invoices

Tableau 2 Table invoices — donnees source (extrait)

invoice_id	company_id	invoice_date	status	doc_type	currency
INV-2024-001	C001	2024-01-15	COMPTABILISE	FACTURE	EUR
INV-2024-002	C001	2024-02-15	COMPTABILISE	FACTURE	EUR
INV-2024-003	C001	2024-03-15	**RECU**	FACTURE	EUR
INV-2024-004	C001	2024-03-15	COMPTABILISE	FACTURE	EUR
INV-2024-005	C002	2024-01-10	COMPTABILISE	FACTURE	EUR
INV-2024-006	C002	2024-02-10	COMPTABILISE	FACTURE	EUR
INV-2024-007	C003	2024-01-20	COMPTABILISE	FACTURE	EUR
INV-2024-008	C004	2024-01-05	COMPTABILISE	**AVOIR**	EUR

INV-2024-009	C005	2024-01-25	COMPTABILISE	FACTURE	**USD**
INV-2024-010	C006	2024-01-30	COMPTABILISE	FACTURE	EUR
INV-2024-011 a 022	C001	2024-04 a 12	COMPTABILISE	FACTURE	EUR
INV-2025-001 a 012	C001	2025-01 a 12	COMPTABILISE	FACTURE	EUR
INV-2026-001	C001	2026-01-15	COMPTABILISE	FACTURE	EUR
INV-2026-002	C001	2026-02-15	COMPTABILISE	FACTURE	EUR

> ***Anomalies identifiees :*** *INV-2024-003 est RECU (provisionnel). INV-2024-004 partage la meme date (doublon de date, transition prov->def). INV-2024-008 est un AVOIR (filtre). INV-2024-009 est en USD (devise etrangere). INV-2024-010 appartient a C006 (nom NULL).*

2.3 Table invoice_lines

Tableau 3 Table invoice_lines — donnees source

line_id	invoice_id	amount	label	category
L001	INV-2024-001	120.00	Eau potable	UTILITIES
L002	INV-2024-002	120.00	Eau potable	UTILITIES
L003	INV-2024-003	120.00	Eau potable	UTILITIES
L004	INV-2024-004	120.00	Eau potable	UTILITIES
L005	INV-2024-005	200.00	Electricite	UTILITIES
L006	INV-2024-006	200.00	Electricite	UTILITIES
L007	INV-2024-007	45.00	Internet fibre	TELECOM
L008	INV-2024-008	**-50.00**	Avoir fournitures	OFFICE
L009	INV-2024-009	5000.00	Consulting IT	SERVICES
L010	INV-2024-010	300.00	**NULL**	NULL
L011-L022	INV-2024-011 a 022	120.00	Eau potable	UTILITIES
L023-L034	INV-2025-001 a 012	120.00	Eau potable	UTILITIES
L035	INV-2026-001	120.00	Eau potable	UTILITIES
L036	INV-2026-002	120.00	Eau potable	UTILITIES

2.4 Table payments

Tableau 4 Table payments — donnees source

payment_id	invoice_id	paid_amount	payment_date	method
P001	INV-2024-001	120.00	2024-01-20	VIREMENT
P002	INV-2024-002	120.00	2024-02-20	VIREMENT
P003	INV-2024-003	**0.00**	**NULL**	NON_PAYE
P004	INV-2024-004	120.00	2024-03-20	VIREMENT
P005	INV-2024-005	200.00	2024-01-15	VIREMENT

3. Configuration du Pipeline

Configuration JDBC (POST /pipelines/{id}/mapping)

```
{
  "jdbcUrl": "jdbc:postgresql://askandgo-db:5432/erp",
  "jdbcUsername": "readonly",
  "jdbcPassword": "***",
  "tables": [
    {"name": "invoices", "alias": "inv"},
    {"name": "companies", "alias": "c"},
    {"name": "invoice_lines", "alias": "l"}
  ],
  "joins": [
    {
      "type": "INNER",
      "fromAlias": "inv",
      "toAlias": "c",
      "condition": "inv.company_id = c.company_id"
    },
    {
      "type": "INNER",
      "fromAlias": "inv",
      "toAlias": "l",
      "condition": "inv.invoice_id = l.invoice_id"
    }
  ],
  "tenantIdColumn": "c.tenant_code",
  "fieldMapping": {
    "supplier": "c.name",
    "amount": "l.amount",
    "date": "inv.invoice_date",
    "label": "l.label"
  },
  "importStatusColumn": "inv.status",
  "importStatuses": ["RECU", "COMPTABILISE"],
  "provisionalStatuses": ["RECU"],
  "finalStatuses": ["COMPTABILISE"],
  "conditions": [
    {"column": "c.tenant_code", "operator": "=", "value": "whitecape_ask"},
    {"column": "inv.doc_type", "operator": "=", "value": "FACTURE"},
    {"column": "inv.currency", "operator": "=", "value": "EUR"},
    {"column": "c.is_active", "operator": "=", "value": "true"},
    {"column": "l.amount", "operator": ">", "value": "0"}
  ],
  "groupCols": ["supplier", "label"],
  "schedule": {"mode": "MANUAL"}
}
```

4. Requete SQL generee

```sql
SELECT
    c.name AS supplier,
    l.amount AS amount,
    inv.invoice_date AS invoice_date,
    l.label AS label
FROM invoices inv
INNER JOIN companies c ON inv.company_id = c.company_id
INNER JOIN invoice_lines l ON inv.invoice_id = l.invoice_id
WHERE c.tenant_code = ?          -- 'whitecape_ask'
  AND inv.doc_type = ?           -- 'FACTURE'
  AND inv.currency = ?           -- 'EUR'
  AND c.is_active = ?            -- true
  AND l.amount > ?               -- 0
  AND inv.status IN (?, ?)       -- 'RECU', 'COMPTABILISE'
```

Les parametres sont injectes via `PreparedStatement` pour eviter les injections SQL. L'ordre des parametres correspond a l'ordre d'apparition dans la clause `WHERE`.

5. Etapes d'import et resultats

5.1 Etape 1 : Extraction JDBC

Tableau 5 Resultat de l'extraction JDBC — lignes filtrees

Ligne brute	Statut	Raison
C007 + donnees associees	**FILTRE**	`tenant_code = 'other_tenant'` != `whitecape_ask`
INV-2024-008 (AVOIR)	**FILTRE**	`doc_type = 'AVOIR'` != `FACTURE`
INV-2024-009 (USD)	**FILTRE**	`currency = 'USD'` != `EUR`
L008 (montant -50)	**FILTRE**	`l.amount > 0` echoue (-50 <= 0)

Resultat : 36 lignes extraites, 4 filtrees par conditions SQL.

5.2 Etape 2 : Normalisation (InvoiceNormalizer)

Tableau 6 Resultat de la normalisation

Ligne	Montant brut	Montant parse	Date parsee	Statut	Raison rejet
C001, 120.00, 2024-01-15	"120.00"	120.0	2024-01-15	VALID	—
C001, 120.00, 2024-02-15	"120.00"	120.0	2024-02-15	VALID	—
C001, 120.00, 2024-03-15 (RECU)	"120.00"	120.0	2024-03-15	VALID	—
C001, 120.00, 2024-03-15 (COMPTABILISE)	"120.00"	120.0	2024-03-15	VALID	—
C006, 300.00, 2024-01-30	"300.00"	300.0	2024-01-30	**REJETE**	`supplier is null or blank`
L010 (label NULL)	"300.00"	300.0	2024-01-30	**REJETE**	`label is null or blank`

Resultat : 34 lignes valides, 2 rejetees (normalisation).

5.3 Etape 3 : Transition Provisionnel a Definitif

Scenario : INV-2024-003 (RECU) et INV-2024-004 (COMPTABILISE)

Meme company C001, meme date 2024-03-15, meme montant 120.00.

Import 1er passage (RECU)

Tableau 7 Etat de la facture provisionnelle

Champ	Valeur
`invoiceId`	INV-2024-003
`status`	RECU
`isFinal`	**false**
`accountingStatus`	"RECEIVED"
`receivedDate`	now()
`amount`	120.00

Import 2eme passage (COMPTABILISE)

```
-- Recherche du provisionnel associe
SELECT * FROM invoices
WHERE tenant_id = ?
  AND pipeline_id = ?
  AND series_id = ?
  AND date = '2024-03-15'
  AND isFinal = false
  AND ignored = false

-- TROUVE : INV-2024-003

-- TRANSITION (mutation du provisionnel) :
UPDATE invoices SET
    amount           = 120.00,
    isFinal          = true,
    accountingStatus = 'VALIDATED',
    validationDate   = now(),
    source           = 'ACCOUNTING_POSTING'
WHERE invoice_id = 'INV-2024-003'
```

Tableau 8 Mapping des champs lors de la transition

Champ	Avant	Apres
`amount`	120.00	120.00 (copie)
`isFinal`	false	**true**
`accountingStatus`	"RECEIVED"	"VALIDATED"
`validationDate`	NULL	now()
`source`	"WORKFLOW_IMPORT"	"ACCOUNTING_POSTING"

> ***Resultat final :*** *1 seule facture en base,* `isFinal=true`*,* `VALIDATED`*. INV-2024-003 est mute, pas de doublon cree.*

5.4 Etape 4 : Detection de doublons

Tableau 9 Tests de doublons

Test	Resultat
INV-2024-003 et INV-2024-004	**PAS DOUBLON** — transition provisionnel a definitif
Memes lignes re-importees	**DOUBLON DETECTE** — `findTopByTenantIdAndPipelineIdAndSupplierAndAmountAndDate` retourne existant

5.5 Etape 5 : Series creees (apres confirmation)

Tableau 10 Series generees par compute_series.py

Serie	Fournisseur	Libelle	N	Mu	Sigma	CV	UseSeasonality	MedianGapDays
S001	EAU_SAISON	Eau potable	24	120.0	0.0	0.0	**false**	~30
S002	ELEC_HEBDO	Electricite	2	200.0	0.0	0.0	**false**	~30
S003	INTERNET_FIBRE	Internet fibre	1	45.0	0.0	0.0	**false**	—

Analyse saisonnalite EAU_SAISON

- Tous les montants = 120.0 (identiques)
- `std_global = 0`
- `avg_monthly_std = 0`

- `ratio = 0/0` → division par zero → **useSeasonality** = **false** (garde-fou dans le code)

6. Anomalies detectees

6.1 Anomalie 1 : Facture avec montant anormal

Import ulterieur : `EAU_SAISON`, `Eau potable`, **500.00 EUR**, `2026-03-15`, `COMPTABILISE`

Parametres de detection

Tableau 11 Parametres detect.py — Anomalie AMOUNT

Parametre	Valeur
`useSeasonality`	false
`mu`	120.0
`tolerancePct`	10.0
`max_acceptable`	132.0
`amount`	500.00

Calcul du score

```
excess      = 500.00 - 132.00 = 368.00
denominator = 120.0 x 10 / 100 = 12.0
score       = min(100, 60 + (368.00 / 12.0) x 25)
score       = min(100, 60 + 30.67 x 25)
score       = min(100, 60 + 766.67)
score       = 100.0
severity    = "CRITIQUE"
```

Resultat

Tableau 12 Alerte creee — Anomalie AMOUNT

Champ	Valeur
`anomalyType`	AMOUNT
`score`	100.0
`severity`	CRITIQUE
`explanation`	"Montant 500.00 depasse la limite 132.00 (normal attendu 120.00, tolerance 10 %)"

`reference_mu`	120.00
`max_acceptable`	132.00

6.2 Anomalie 2 : Facture manquante (apres delai)

Etat serie EAU_SAISON :

- `lastValidatedDate` = 2026-02-15
- `medianGapDays` = 30
- `toleranceDays` = 10

Condition d'alerte (aujourd'hui = 2026-03-30)

```
nextExpected = 2026-02-15 + 30j = 2026-03-17
deadline     = 2026-03-17 + 10j = 2026-03-27
today (2026-03-30) > deadline (2026-03-27) --> ALERTE
```

Tableau 13 Alerte creee — Anomalie MISSING

Champ	Valeur
`anomalyType`	MISSING
`score`	100.0
`severity`	CRITIQUE
`explanation`	"Facture manquante : attendue vers le 2026-03-17 (dernier recu le 2026-02-15, delai habituel 30j)"
`expectedDate`	2026-03-17

7. Feature F — Budget (Serie EAU_SAISON)

Historique (2 annees completes)

Tableau 14 Donnees historiques par annee — EAU_SAISON

Annee	Mois 1	Mois 2	Mois 3	...	Mois 12	Total
2024	120	120	120	...	120	**1440**
2025	120	120	120	...	120	**1440**

Budget implicite 2026 = (1440 + 1440) / 2 = **1440.0 EUR**

Repartition mensuelle (non saisonnier, mu = 120, 12 mois)

Tableau 15 Budget mensuel 2026 — repartition lineaire

Mois	Attendu (EUR)	Realise (fév 2026)	Statut
1	120.00	120.00	NORMAL
2	120.00	120.00	NORMAL
3	120.00	—	UPCOMING
4	120.00	—	UPCOMING
5	120.00	—	UPCOMING
6	120.00	—	UPCOMING
7-12	120.00	—	UPCOMING
Total	**1440.00**	**240.00**	

> *Puisque `useSeasonality = false`, le budget est reparti uniformement sur 12 mois (120 EUR/mois). Avec `useSeasonality = true`, la repartition suivrait les `monthly_mu`.*

8. Tableau recapitulatif des anomalies

Tableau 16 Synthese des anomalies detectees

#	Type	Declencheur	Score	Severite	Action utilisateur
1	`AMOUNT`	Facture 500 EUR (mu = 120 EUR)	100	CRITIQUE	CONFIRMED / REJECTED / IGNORED
2	`MISSING`	Aucune facture apres 40 jours	100	CRITIQUE	CONFIRMED / REJECTED / IGNORED
3	`DUPLICATE`	Re-import identique	100	CRITIQUE	Auto-detecte, pas d'alerte si filtre

9. Pipeline Run Logs

Tableau 17 Journal d'execution du pipeline (extrait)

Etape	Statut	Message	Lignes concernees
JDBC Fetch	SUCCESS	36 lignes extraites	—
Condition Filter	SKIPPED	4 lignes filtrees (tenant, type, devise, montant)	C007, INV-008, INV-009, L008
Normalization	FAILED	2 lignes rejetees	C006 (supplier null), L010 (label null)
Provisional a Final	SUCCESS	1 transition	INV-003 --> INV-004
Duplicate Check	SKIPPED	0 doublons	—
Scoring	ALERT	1 anomalie AMOUNT	Facture 500 EUR
Missing Check	ALERT	1 anomalie MISSING	Delai depasse

10. Resume des comportements cles (JDBC)

Tableau 18 Synthese des comportements attendus — Pipeline JDBC

Feature	Comportement avec donnees JDBC multi-tables
Joins SQL	3 tables jointes, conditions structurees, PreparedStatement
Filtrage tenant	`tenantIdColumn` injecte `c.tenant_code = ?`
Filtrage statuts	`importStatuses` limite a RECU / COMPTABILISE
Filtrage conditions	doc_type = FACTURE, currency = EUR, amount > 0, is_active = true
Normalisation	Rejet si supplier/label null, parsing montant/date
Transition prov a def	Matching par (serie, date), mutation du provisionnel
Doublons	Detection (supplier, amount, date), batch + DB
Saisonnalite	false (donnees constantes), true (si variation mensuelle)
Budget	Moyenne N-1 / N-2, scaling, repartition mensuelle
Alertes	AMOUNT (score > 60), MISSING (delai depasse), BUDGET_* (seuils)
Feedback	CONFIRMED / REJECTED / IGNORED — ajuste tolerance / exclusion